The Halving and its impact on hash rate and miners cost structures — 2024 Mining Report

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Executive Summary

- **Growth and Sustainability Concerns:** The Bitcoin mining network has grown by 90% in 2023, raising questions about its environmental sustainability and profitability, particularly the efficiency and energy cost of the network.

- **Mining Dynamics and Hashrate Trends:** Bitcoin mining's "difficulty" adjustment mechanism ensures

supply inelasticity. Post-halving, higher-cost miners may struggle due to decreased immediate income. This document assesses the average cost of production per Bitcoin post-halving, with findings showing an average cost of US$37,856.

- **Efficiency Improvements in the Mining Network:** Despite the increasing power demand of the network, there have been significant improvements in efficiency. Using nonce data has helped analyse different mining models' efficiency. Currently, the network's average efficiency is 34W/T, with projections showing a potential drop to 10W/T by mid-2026.

- **Bitcoin Mining and Environmental Impact:** Bitcoin mining often utilises stranded energy, typically in remote locations. Approximately 53% of Bitcoin mining energy is now sustainably sourced according to Daniel Batten. Bitcoin mining could significantly reduce emissions caused by gas flaring, a major environmental concern.

- **Financial Analysis of Miners Post-Halving:** The cost of production and profitability structures for miners will change following the 2024 halving. Our analysis focuses on the varying cost structures of listed miners and their vulnerabilities to the halving.

- **Conclusion and Miner Positioning:** Most miners will face challenges from high SG&A costs, necessitating cost reductions to remain profitable. Only a handful of miners are expected to operate profitably if Bitcoin prices remain above $40,000.

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The Bitcoin Mining Network Growth

The Bitcoin mining network has experienced meteoric growth, with an increase in hashrate of 104% in 2023. This rapid expansion raises concerns regarding its sustainability, both from an environmental standpoint and in terms of the mining network's profitability. In this document, we aim to address both of these important issues. Our final results of the average cost of production per Bitcoin, post halving for each miner are listed below, highlighting the average cost of production is US$37,856.

While Bitcoin mining shares some parallels with traditional mining, in that energy is expended to yield an asset of value, the similarities largely stop there. The unique self-regulating mechanism known as the “difficulty”

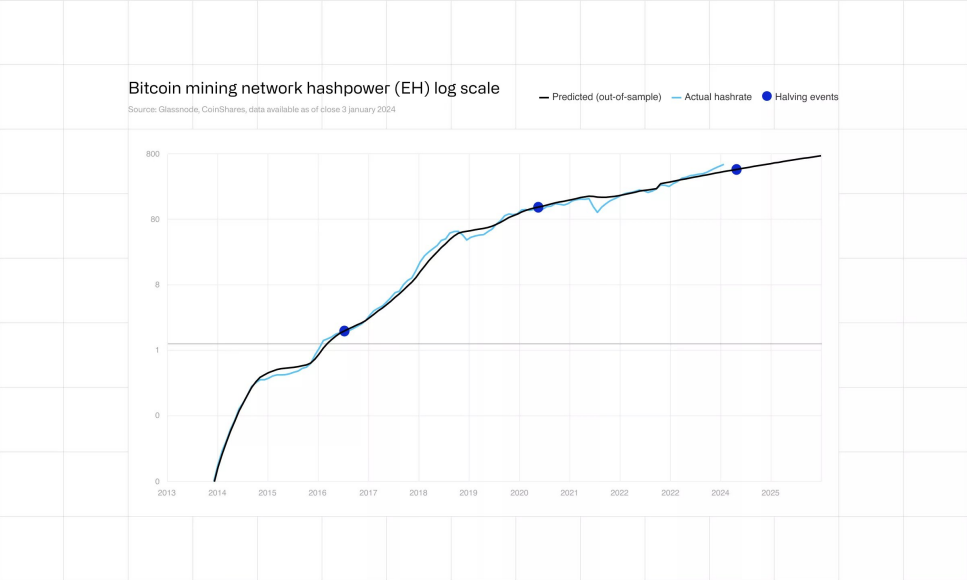
adjustment in Bitcoin mining ensures that supply remains strictly inelastic. At a certain point in the cycle of bitcoin mining, miners at the higher end of the cost curve will begin to suffer, and the hashrate will begin to decline as price rises aren't enough to offset the rising difficulty in mining.

Regardless of whether there are 2 or 2 million miners, the amount of new Bitcoin created remains constant until the next scheduled halving event. If the collective hashpower of the network increases significantly, the mining difficulty will adjust upwards to keep production rates on target, consequently pushing higher-cost miners out of the market. Our analysis focuses on the differing cost structures of listed miners, and those most vulnerable to the impacts of the halving in April 2024.

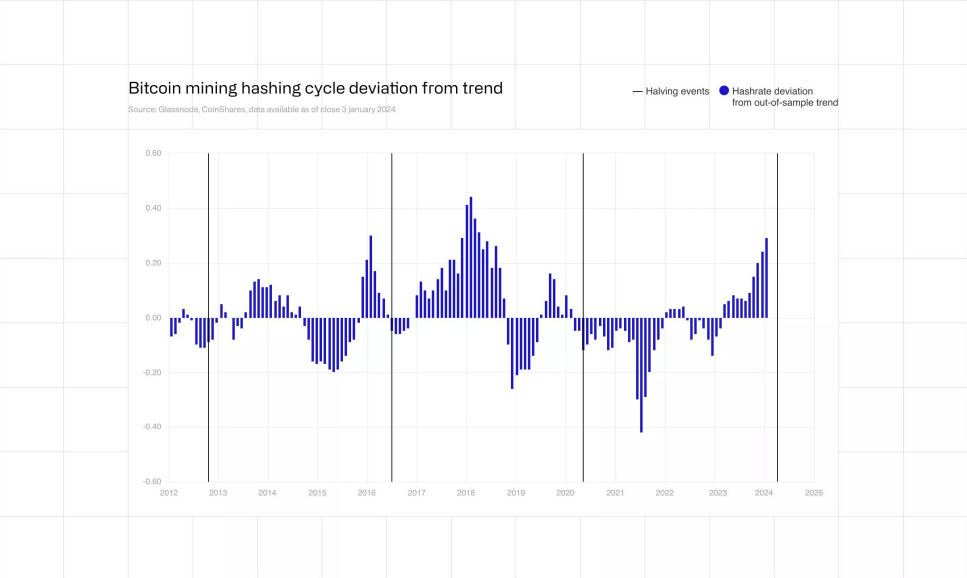
To forecast the future direction of the hashrate, our best method is to analyse historical patterns. A qualitative reasoning suggests that growth will be partially driven by Bitcoin's price: a positive growth outlook could encourage miners to increase the hashrate, perceiving it as profitable. However, this depends on assumptions about future prices.

A review of historical data indicates that mining activity has risen between halving events. Yet, due to its exponential growth, identifying a clear cycle is challenging - we have done some work on this here. As the hashrate tends to be a volatile figure, using deviation from the trend in the historical data yields more accurate results than purely qualitative methods. A key issue, however, is that most trend lines incorporate future information, meaning the trend line we see today would have appeared differently in

the past. Therefore, the most reliable method is to calculate a trend line that is based on out-of-sample data, which isn't influenced by future developments.



The data reveals an intriguingly regular pattern occurring between the halving cycles, indicating that the current peaks in hashrate are not unusual from a historical standpoint for this point in the cycle. This trend is more clearly illustrated in the chart below, which distinctly showcases the cyclical nature of these changes.



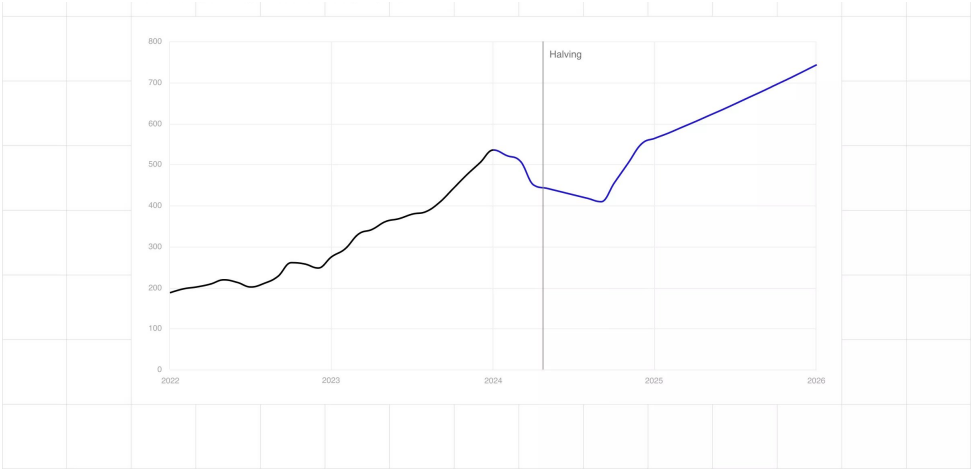
Since the first Bitcoin halving in 2012, followed by

subsequent halvings in 2016 and 2020, a pattern has emerged where the hashrate typically drops about 9% below the trend line post-halving, a situation that usually lasts for around six months. The year 2020 was somewhat exceptional, as this period extended considerably due to China's ban on mining, leading to a 42% drop below the trend line. Nonetheless, the pattern generally involves an initial decline in hashrate, followed by a recovery midway through the cycle, and then a surge in activity approximately one year before the next halving.

This cycle is logical: To stay competitive in anticipation of the halving, miners increase their capital expenditure, driving the hashrate significantly above the trend. The miners earn less immediate income after a halving, impacting their capital expenditure cycles. The current cycle is no different. Notably, the peak in hashrate growth often occurs about four months before the halving, likely due to a "Bitcoin rush" that leads to a spike in mining difficulty, which in turn pushes out miners and mining rigs with higher costs of production. The current mining difficulty is at historical highs, and aligns with “relative” peaks observed in previous cycles.

What does the future hold for Bitcoin's hashrate? Using historical trends as a guide, we might expect the hashrate to normalise back to the trend line at around 450EH/s (exahash per second) by the April 2024 halving. It could potentially decrease further to 410EH/s six months later. Following that, the trend line forecasts a sharp increase in the hashrate to approximately 550EH/s by the end of 2024.



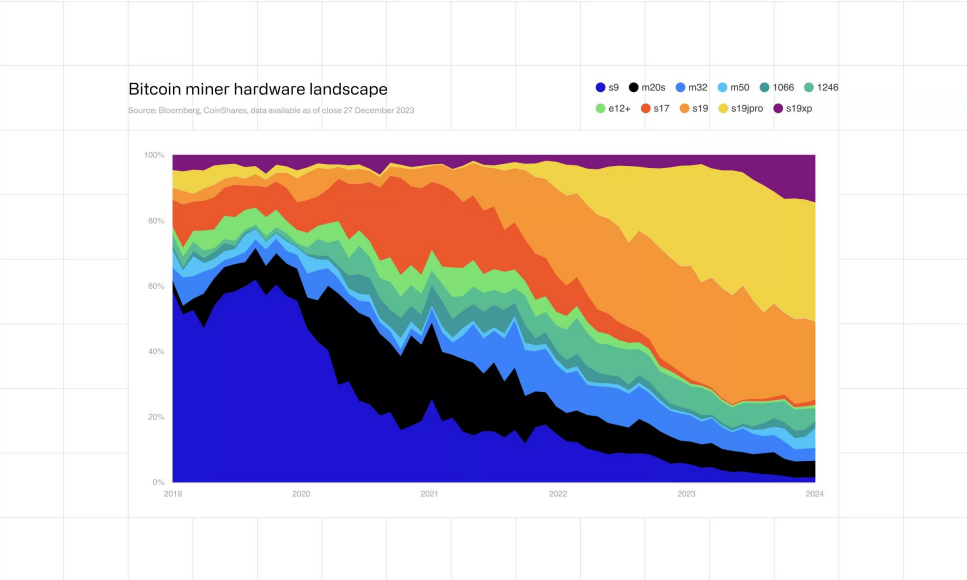


This halving is likely to kick out those along the higher end of the cost curve, leaving those that remain who have ample liquidity with a great opportunity to acquire hardware at a discount. This scenario depends very much on whether the price rises above the average cost of production for each miner, and would likely require either a significant price drop, or a large drop in transaction fees, such as a decline in Ordinal usage.

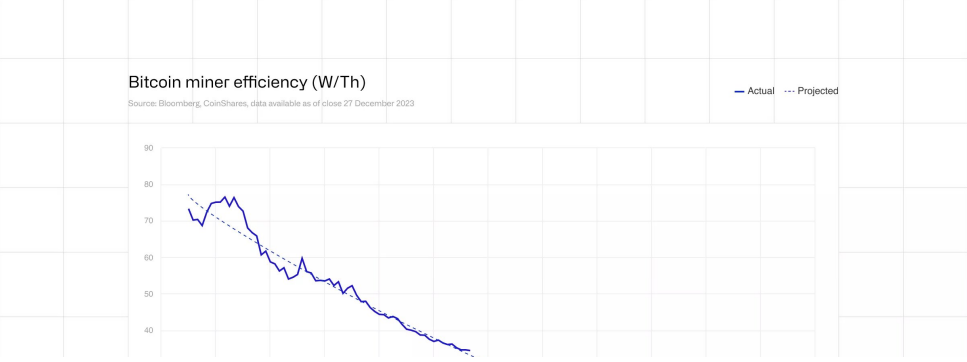
The Improving Efficiency of the Mining Network

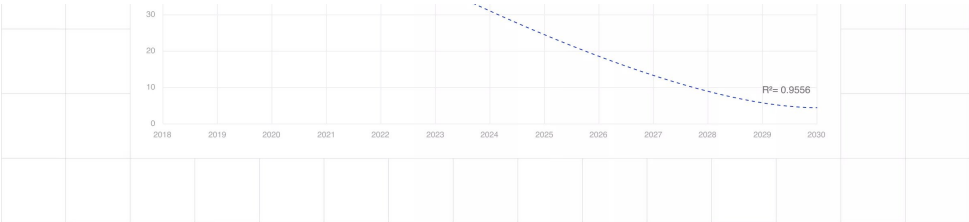
The array of mining equipment currently used for Bitcoin is diverse, encompassing a range of power consumption levels, hashpower, and resulting efficiencies. Historically, this diversity has made it challenging to determine the overall efficiency of the mining fleet. Karim Helmy from CoinMetrics has conducted some notable research using nonce data for hardware fingerprinting. To avoid delving too deeply into technicalities, it was found that each miner model leaves a distinct 'vapour trail' on the Bitcoin

blockchain. This unique signature can then be analysed to ascertain the distribution of different mining models within the network.



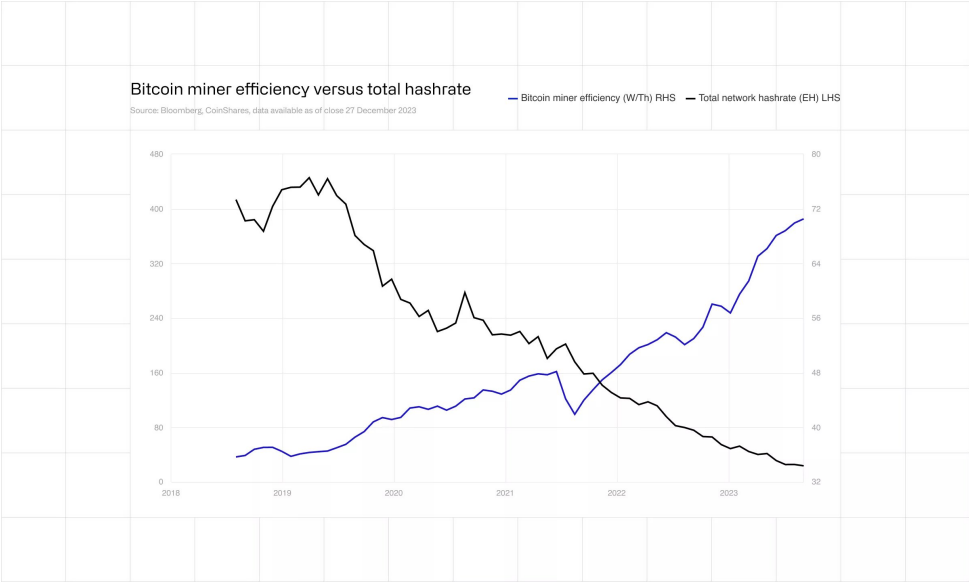
Since the efficiency of each mining model is known in terms of W/T (watts per terahash), it is possible to calculate the overall efficiency of the entire Bitcoin mining fleet. Given the fairly linear progression of this path, future trends can also be projected. Currently, the network boasts a weighted average efficiency of 34W/T. This year alone, there has been an 8% improvement in efficiency, and over the past three years, efficiency was enhanced by 28%. Based on these trends, it is projected that by mid-2026, the efficiency level could reach as low as 10W/T as chip design continually improves and more efficient mining hardware is brought online.



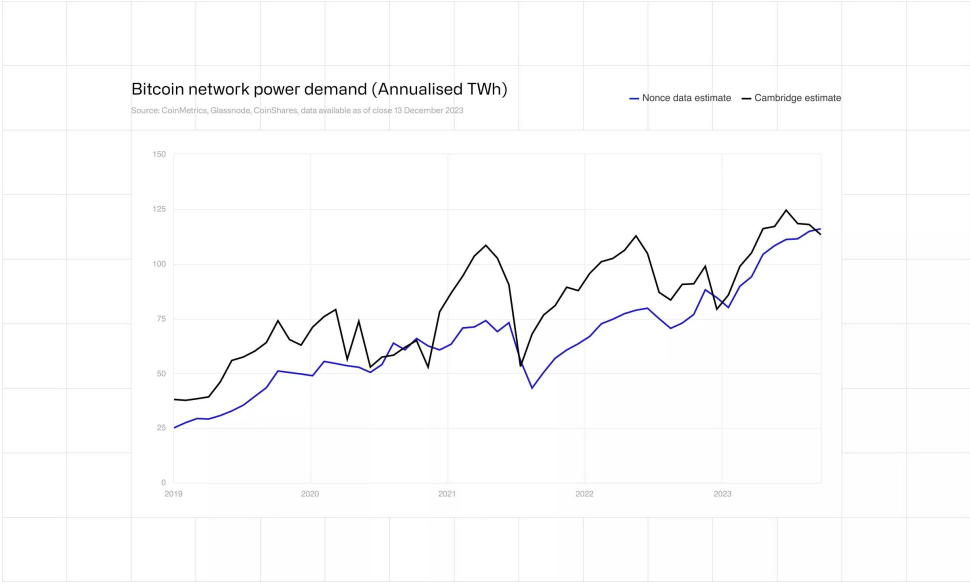


Bitcoin mining consistently pursues the most affordable energy sources, which frequently leads to the utilisation of stranded energy—energy that cannot be easily sold to the existing power grid. Often, this involves renewable energy projects situated in remote locations. As a result, there is a growing trend of Bitcoin mining operations using electricity from sustainable sources. According to estimates from Daniel Batten, approximately 53% of the energy used for mining Bitcoin is now sourced sustainably. This proportion has surpassed that of the finance industry, where, as Daniel Batten notes, only about 40% of energy consumption is estimated to come from sustainable sources.

While the hashrate has risen significantly recently, the efficiency of the network, in stark contrast, continues to fall (i.e., improve).



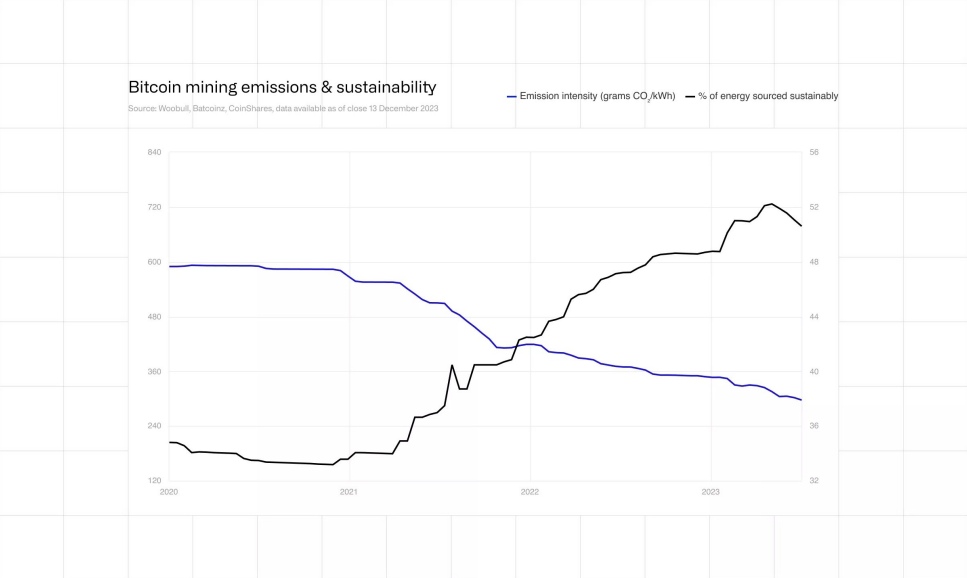
The new level of detail of the CoinMetrics nonce data means that we can estimate the annual power costs, which are remarkably close to estimates made by Cambridge University.



The data underscores that although there have been significant improvements in efficiency, the network's power demand has reached an all-time high of 115 terawatt-hours (TWh) on an annualised basis, marking a 44% increase this year. However, this increase is relatively modest compared to the growth in hashrate, thanks to the ongoing efficiency improvements.

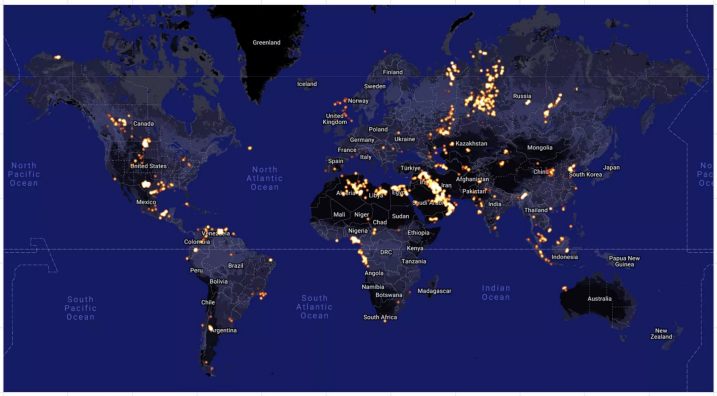
Daniel Batten's research into the mining industry's emission intensity reveals a notable decrease in emissions, although some of the data sources used are difficult to track. Since 2021, emissions have dropped from nearly 600 grams of CO2 per kWh to just 299 grams of CO2 per kWh. This reduction is likely attributed to the substantial increase in the use of sustainably sourced energy, which has grown from 33% in 2021 to 52% today. This is mirrored to some extent by the ERCOT (Texas) grid fuel mix, where a significant proportion of Bitcoin mining occurs, seeing

renewable energy grow from 20% of total energy produced in 2017, to 31% in 2023 according to data from IEEFA.

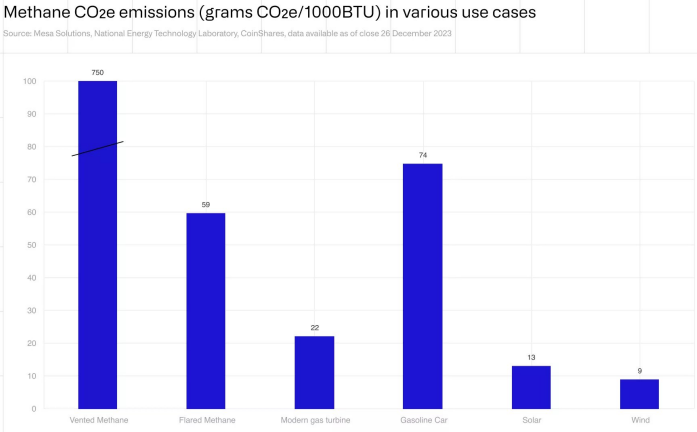


Bitcoin can contribute to a reduction in carbon emissions

Gas flaring is becoming an increasingly serious problem, as underscored in a recent BBC report. This report brought attention to the fact that oil drilling activities in the Gulf, along with the associated practice of flaring excess gas, are presenting a more significant threat to millions of people than previously understood. While flaring is environmentally preferable to venting, as it reduces CO2 equivalent emissions by 92% according to Mesa Solutions, its widespread use remains a concern. The image from SkyTruth strikingly showcases the extent of this global issue, with yellow dots vividly marking areas of flaring activity.



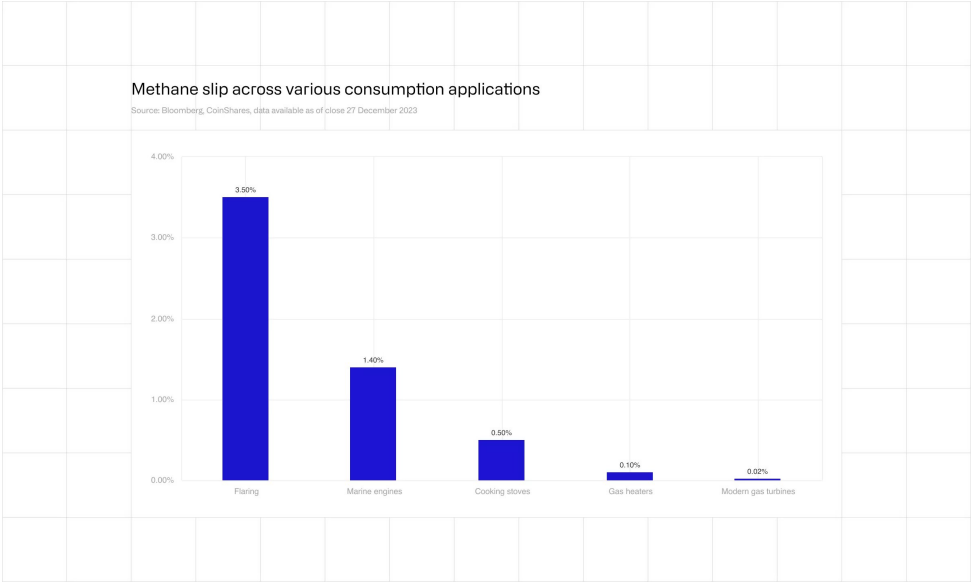
The World Bank estimated that in 2022 approximately 139 billion cubic metres of natural gas were flared worldwide. This quantity is comparable to the combined total gas consumption of Central and South America. Presently, the conventional practice of flaring methane results in the emission of 59 grams of CO2 equivalent (CO2e) per 1000 British Thermal Units (BTU), as per data from Mesa Solutions. In contrast, utilising a modern turbine electricity generator would only emit 22 grams of CO2e per 1000 BTU. This represents a 63% reduction in emissions, making it three times less polluting than a gasoline-powered car.



The main challenge with flaring lies in the fact that it

involves energy which cannot be economically stored or transported and, therefore, is often burned off. This typically happens in remote locations where connecting to power grids or pipelines isn't practical. We believe that Bitcoin mining could significantly contribute to reducing emissions caused by flaring. This is because mining hardware, along with the necessary generators, can be housed in containers and operated in these remote areas, far from established power grids.

Additionally, flaring often results in a higher incidence of methane slip. This phenomenon happens when a small portion of natural gas fails to combust completely, consequently escaping into the atmosphere, a situation especially prevalent in windy conditions. In contrast, turbines are known to have one of the lowest rates of methane slip, significantly minimising the risk of such occurrences.



At present, gas flaring contributes to approximately 406 million tonnes of CO2 emissions each year. However, if all the gas currently being flared were instead used for Bitcoin mining, these emissions could potentially be reduced to

about 152 million tonnes of CO2. As global flaring currently comprises 1.1% of global CO2 emissions, bitcoin mining could reduce global flaring emissions to just 0.41% of global emissions.

As of now, only around 120 megawatts (MW) of Bitcoin mining capacity is known to be harnessing energy from stranded gas. Therefore, Bitcoin mining holds significant potential to markedly decrease global emissions, should it expand its use of this otherwise wasted flared gas.

The Impact of the Halving on Bitcoin Miners

In this research article, we have estimated the weighted average for the cost of production and cash cost, which stood at approximately $16,800 and $25,000 per bitcoin respectively for Q3 2023. Following the halving event, expected to happen in April 2024, these costs are likely to rise to $27,900 and $37,800 respectively. Riot looks to be best positioned to navigate these changes due to their efficient cost structures and long runway. Our analysis of the listed and private miner financial statements assumes a Bitcoin price of $40,000, with most of the coming pain for miners likely stemming from bloated Selling, General, and Administrative Expenses (SG&A) costs.

Methodology

Our approach to financial analysis for Q3 2023 involved working out an adjusted consolidated income statement. This standardisation was applied to the mining operations of 14 miners, with 13 of them being publicly listed entities, representing 19% of all Bitcoin mining by hashpower, as of December 2023, and ~25% post-halving with a network hashrate of 450 EH/s. Our data for Q3 2023 primarily derives from publicly available information found in SEC Filings, website production reports, or estimates where necessary.

Our methodology comprises:

- Cost of Revenues represent the self-mining cost of revenues, mostly electricity cost.

- SG&A eliminates non-cash expenses, such as stock-based compensation, one-time payments.

- Interest Expense considers only interest on debt, excluding lease expenses or other finance costs.

Following the halving event, the direct cost of production and operating breakevens change dramatically to $27,900 and $37,800. This change is a function of the halved block subsidy. Our methodology to project income and expenses is as follows:

- Hashrate Market Share is determined by the projected hashrate disclosed by each company, expressed as a percentage of our estimated 450 network EH/s at the halving, using an approximate 10% decline from 500EH/s.

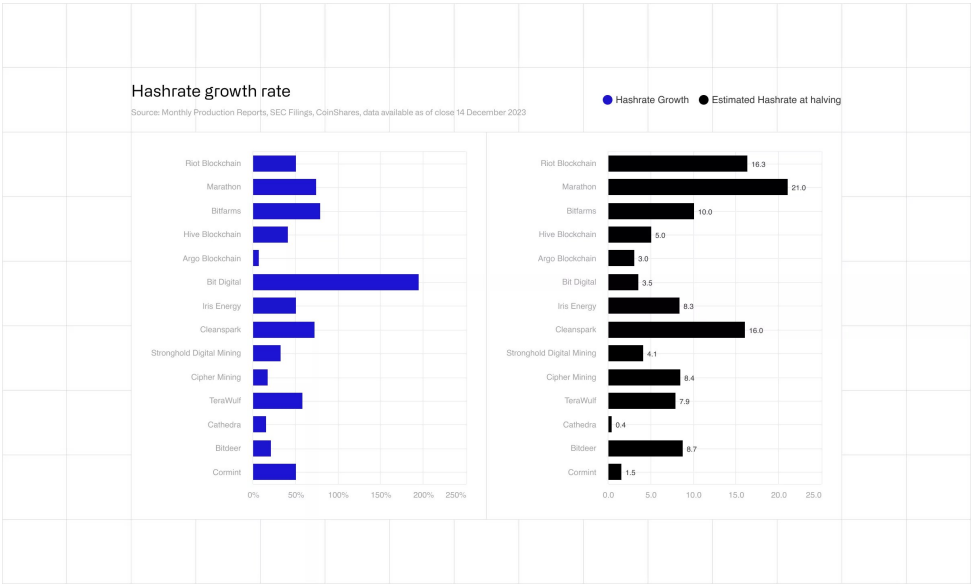
- Cost of production is calculated using a bottom up method of fleet efficiency, cost of electricity, kWh utilised (taking flat Q3 2023 utilisation rates) and number of bitcoin mined.

- SG&A expenses are assumed to remain consistent with Q3 2023, as corporate expenses are not expected to significantly change due to the halving.

- Interest expense is determined as the sum of outstanding principal amounts multiplied by periodic interest rates.

- Our comprehensive methodology ensures a standardised and comparable financial analysis for both pre and post-halving scenarios, guiding our operational decisions effectively.

Miner Computing Power



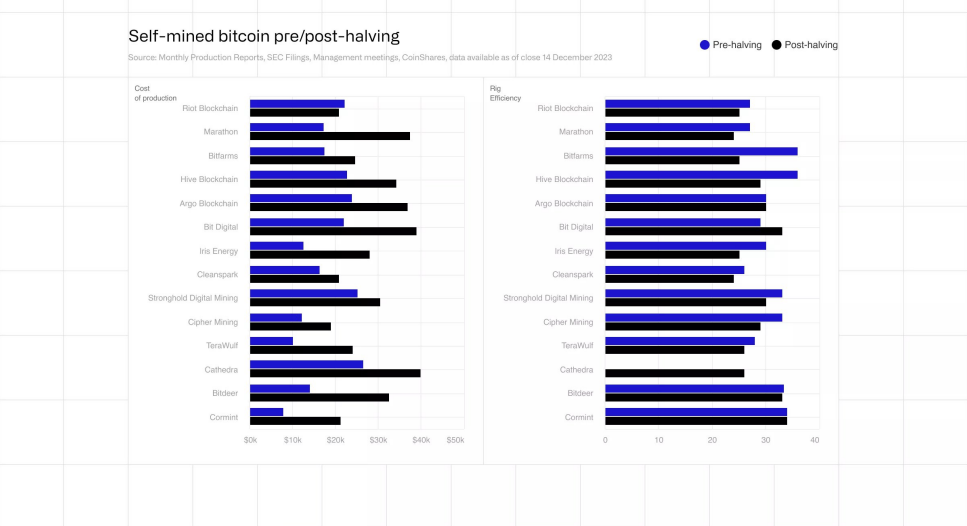
Every ten minutes on average, the Bitcoin network produces a block by a miner correctly computing ('guessing') the hash (a pseudo-random 64 digit

alphanumeric number) with the rest of the network verifying its correctness. A miner with more hashrate (more rigs, leading to more computing power) commands a larger percentage of the network hashrate, and hence has a larger chance of creating a block and earning the block subsidy (currently 6.25 Bitcoin, but will cut in half at 3.125 Bitcoin around April 2024, because of the 'Halving') plus transaction fees. Miners essentially engage in an arms race to buy and plug in as many machines as possible.

The more rigs a miner has for self-mining, the larger the data centre needed on a megawatt basis. This large capital expenditure is either funded by cash, equity or debt, of which the latter can hurt miners' all-in cost of production due to higher interest expense and put them at risk during bitcoin downturns. This is clear in the case of Core Scientific which went into Chapter 11 at the end of 2022, or Mawson, that failed to make payments on their Marshall Loan, for example, as per their Q3 2023 filings (these examples are not exhaustive).

Number of Bitcoin Produced

The number of bitcoin produced is integral to the unit economics and cost structure of each miner. For miners to achieve their same pre-halving bitcoin output, they would need to double their market share, which is incredibly challenging given the network hashrate growth of ~53% CAGR over the last three years, or growth in the amount of fees collected per block would need to fully make up for the reduction in the block subsidy caused by the halving.



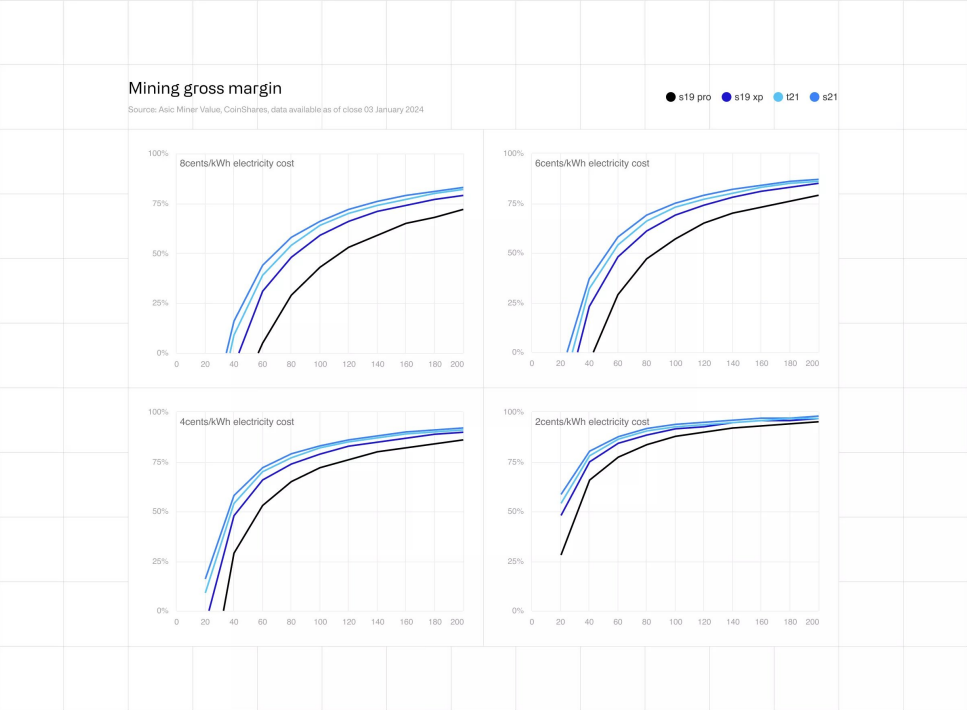
Cost of Electricity

The Bitcoin miners' cost structure is a function of two inputs: energy and equipment. The public miners we track consume a weighted average of 4.5cents per kwh of energy. This energy is purchased in wholesale markets, most likely in the spot or futures market, or negotiated with energy providers via power purchase agreement (PPA) contracts, which usually provide a fixed energy price, but also generally involve take-or-pay clauses. Miners have more control over their fleet of machines, however, and they are able to reduce their energy bills by investing in more efficient machines which consume less power per each hash.

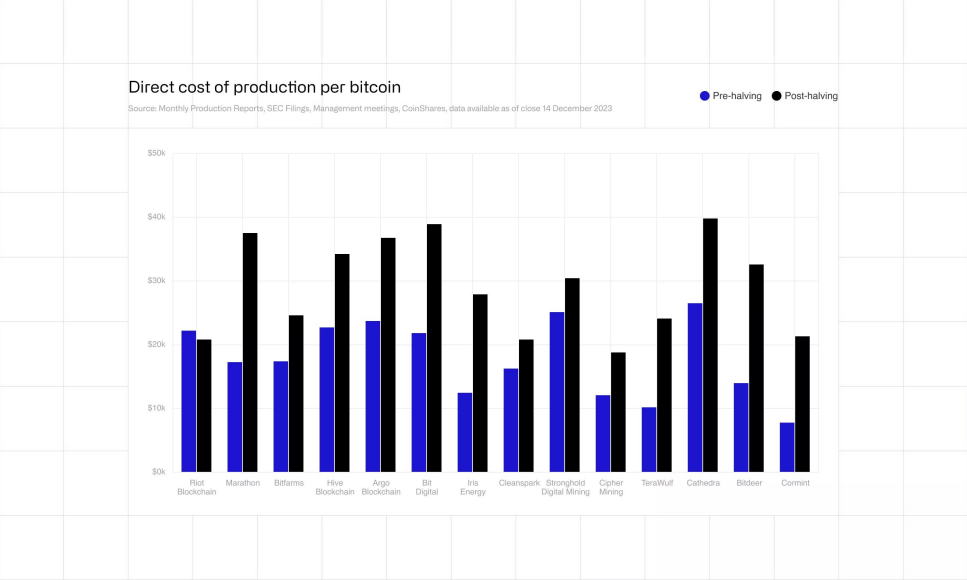
Of the public miners we track, fleet wide efficiency is also expected to decrease from 29W/T to 26W/T at the halving. An example of how miners can upgrade their fleet to improve their efficiency (i.e., reduce their W/T) can be seen at CleanSpark's and Iris Energy's latest machine deals, in which each have purchased 4.4EH/s and 1.4EH/s, respectively, of the Bitmain Antminer S21 miners, with

efficiency ratios of 17.5W/T, at a price of around US$14/TH (US$ per terahash).

The matrix below demonstrates that the S21s outperform all other rig types in every electricity and hashprice scenario, despite that the T21s are a newer model, because of higher hashrate (affecting revenue) and lower power consumption (affecting expenses).



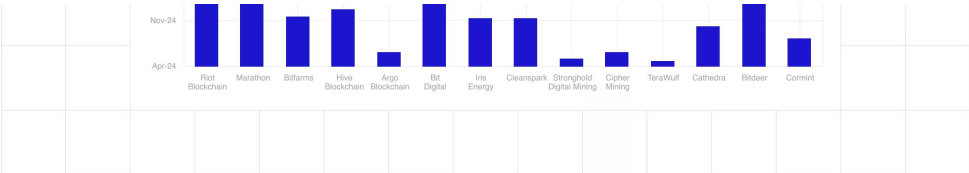
The chart below demonstrates the changes in fleet efficiency pre- and post-halving. Notice that, while most miners are improving their fleet wide efficiency (W/T), their direct cost structure is not improving. This is because, as mentioned before, miners will need to increase their power draw and energy consumed to mine the same amount of bitcoin. Pre- and post-halving, electricity costs per bitcoin took up about 68% and 71% on a weighted average, respectively, of the cash-cost structure, with the slight increase mostly being attributed to scale and slightly higher energy prices.



Miner runway

We define runway as how many days miners can survive paying off their cash operational expenditure using their cash and bitcoin reserves. There is no standard treasury management strategy throughout the industry. Some accumulate as much of their output as possible, also known as ‘HODL’, while others don’t and choose to sell their bitcoin as they are produced. Well capitalised miners with larger bitcoin balances will likely have a higher equity premium in a bull market, such as Riot. However, combining low runway with high cash-cost per bitcoin, exposes miners such as Stronghold to the perils of a low bitcoin price.

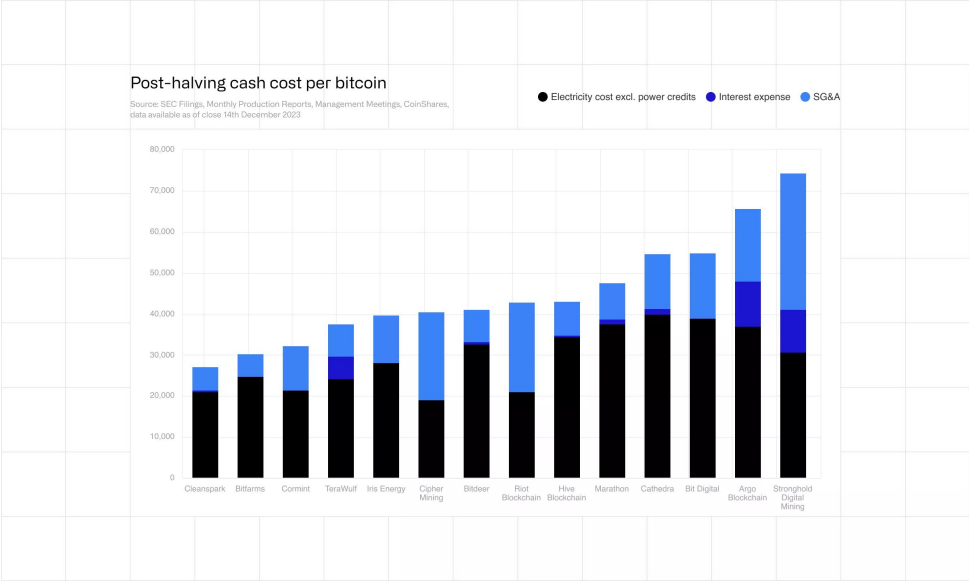




Cash-Cost of Production

As per the figures below, we think Riot, TeraWulf and Cleanspark are best positioned going into the halving. One of the main problems miners have is large SG&A costs. For miners to break even, the halving will likely force them to cut SG&A costs, otherwise they could continue to run at an operating loss and having to resort to liquidating their HODL balances and other current assets.

NB: Iris Energy and Cormint numbers based on Q2 2023 Filings, management meetings and monthly production reports

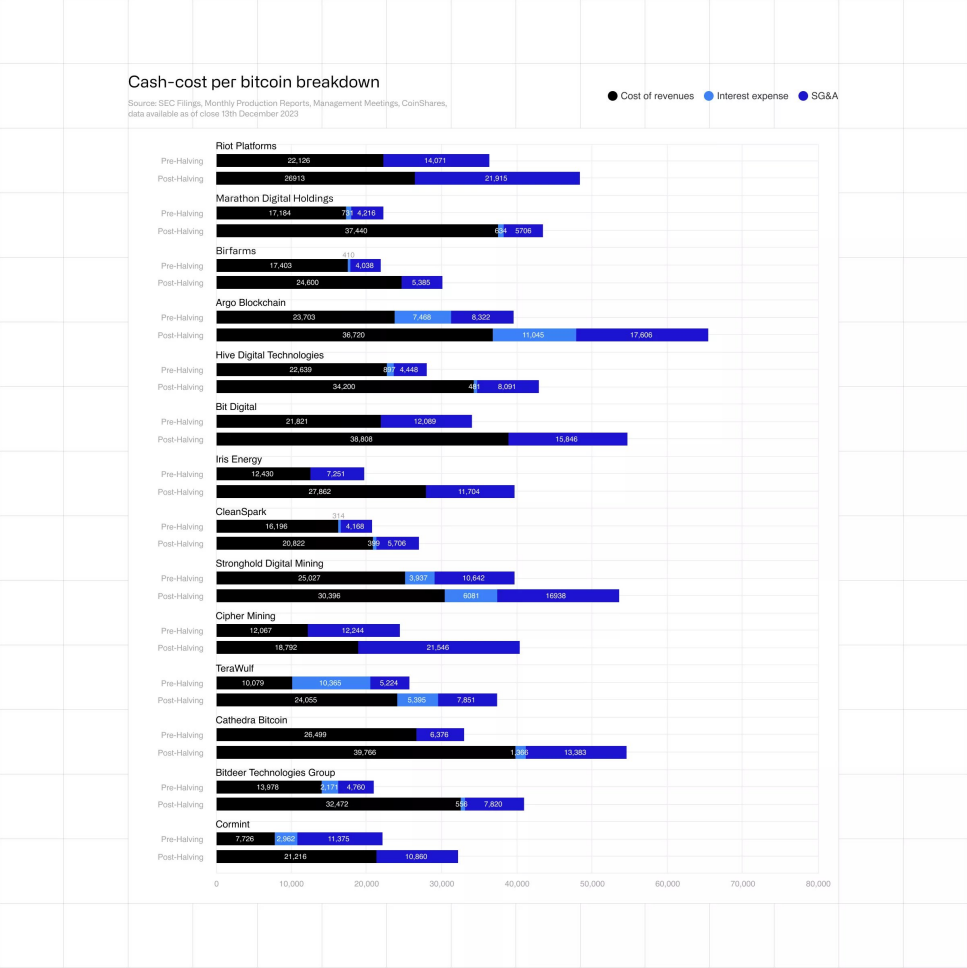


Conclusion

Our analysis suggests that Riot appears to be the best positioned to navigate the complexities of the halving event, chiefly due to their cost structure and long runway. Most of the pain miners will experience likely stems from hefty SG&A expenses which will likely need to be cut to remain profitable.

Overall, unless the price of bitcoin remains above $40,000, we believe that only Bitfarms, Iris, CleanSpark, TeraWulf and Cormint will continue to operate profitably. All the other miners will likely eat into their runway, eventually forcing further dilution of stock prices as they most likely raise equity or convert debt.

Cash-Cost Per Bitcoin Breakdown



NB: Iris Energy based on Q2 filings ; Cormint based on Q2 numbers

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